

Lab 6: Security in the Cloud

I. S3 Bucket Encryption

1. Create an S3 Bucket with Server-Side Encryption
2. Create an S3 bucket with the name *my-secure-bucket* in a region of your choice.
3. Enable Server-Side Encryption with AWS Key Management Service (KMS) and use the default AWS-managed key for encryption.
4. Upload a file to the bucket and verify that the file is encrypted.
5. Configure S3 to automatically encrypt all new files uploaded to the bucket

Submission:

- Screenshot important steps that you have done to get the output
- Explain the pros and cons of using AWS-managed keys versus customer-managed keys.
- How would you configure S3 to automatically encrypt all new files uploaded to the bucket.

II. Launch an RDS Instance with Encryption

1. Launch an **RDS** instance using a supported database engine (e.g., MySQL or PostgreSQL).
2. Enable encryption using the default AWS-managed KMS key during the RDS instance creation process. (**encryption is enable by default**)
3. Verify the encryption status by reviewing the **Configuration** section of the RDS instance in the RDS console.

Submission:

- Screenshot important steps that you have done to get the output

III. Encrypt an Existing RDS Instance

1. Create an RDS instance with encryption disabled.
2. Take a snapshot of an existing RDS instance (ensure it is unencrypted).
3. Copy the snapshot and enable encryption using a KMS key during the copy process.
4. Restore the encrypted snapshot to create a new RDS instance with encryption enabled.

Submission:

- Screenshot important steps that you have done to get the output

- Can you encrypt an existing RDS instance directly without taking a snapshot?
Explain why or why not?
- Does RDS encryption with KMS automatically protect the data in transit between the RDS instance and client applications? Why or why not?

IV. Key Management with AWS KMS

1. Create a new **KMS key** in the **AWS Key Management Service**.
2. Set up key policies and specify IAM users or roles that can access the key.
3. Use the key to encrypt a test file via the **AWS CLI** or SDK.
4. Confirm that the original content of the file is successfully decrypted.

Submission:

- Screenshot important steps that you have done to get the output
- What is encryption and why do we need it?

Remember to remove all services you used.